

*Epiphany is not about solving a complex puzzle, but
something that was too simple to see.*

Paper and Thesis

Keyao Peng. Sheaves and differential equations: An introduction to algebraic analysis, bachelor dissertation. 2019.

Keyao Peng. Milnor-witt motivic cohomology of complements of hyperplane arrangements. *Algebraic & Geometric Topology*, 23(8):3531–3552, 2023.

Keyao Peng. Milnor-witt motivic cohomology and linear algebraic groups. *Advances in Mathematics*, 458:109973, 2024.

Keyao Peng. Milnor-witt motivic decomposition of stiefel varieties. 2024.

Keyao Peng (joint with Haoyang Liu). Cellular $\mathbf{A}^1$ -homology of smooth toric varieties. 2025.

Work Experience

2024–2027 **Post-doc**, CNRS, Institut de Mathématiques de Bourgogne(IMB), Dijon, France

Education

2020–2023 **Ph.D.**, Institut Fourier, Université Grenoble Alpes, Grenoble, France
Algebraic geometry, with advisor *Jean Fasel*

2019–2020 **Master**, Institut Fourier, Université Grenoble Alpes, Grenoble, France
Mathématiques fondamentales

2015–2019 **Bachelor**, Taishan College, Shandong University, Jinan, China
Majored in mathematics

Academic Activities

Speaker

2020–2021 Working group on stratified homotopy theory *IAS, Princeton*

My talk: *Oriented pushouts and oriented fibre products*

2021–2023 Séminaire Compréhensible *Institut Fourier*

My talks:

Feb 2021 *Homotopy type theory for mathematicians*

May 2023 *How to explain (higher) categories to a geometric topologist? (An introduction to Cobordism Hypothesis)*

2022 Géométrie réelle, motifs et $\mathbf{A}^1$ -homotopie

ENS de Lyon

My talk: *Théorie des 6 foncteurs (Theory of six functors)*

Aug 2023	Chow-Witt Rings: Computations and Applications	BIMSA
My talk:	<i>MW-motivic cohomology of linear algebraic groups and Stiefel varieties</i>	
2024-2025	Séminaire Doctorant	IMB
My talks:		
April 2024	<i>0=1-1=-1+1=0: From Elementary School to Higher Algebras</i>	
May 2025	<i>How to build a computer with the string diagrams (Introducing lambda calculus and its visualisation)</i>	
Nov 2024	Motivic homotopy in interaction	CIRM
My talk:	<i>Cellular A1-homology of toric surfaces</i>	
July 2025	Motive and Arithmetic Geometry	Darmstadt
My talk:	<i>Cellular A1-homology of smooth toric varieties</i>	

Teaching

Autumn 2022	TA , Université Grenoble Alpes, An introduction to algebraic geometry
Autumn 2025	Lecture , Université Bourgogne Europe, Introduction to algebraic geometry

Master Supervision

2024-2025	Ragul Nagarajan , Université Bourgogne Europe, Introduction to Moduli Spaces, (joint with Daniele Faenzi)
2025-2026	Sridhar Hariharakrishnan , Université Bourgogne Europe, Six-functor formalism in motivic homotopy theory, (joint with Adrien Dubouloz)
2025-2026	M Rey Georaldo , Université Bourgogne Europe, Differential cohomology and higher structures in differential geometry

Skill

Language

Chinese	Mother tongue	English	C1
French	B2	German	A2
Japanese	A2		

Computer

Language	Typescript, C#, Haskell, Purescript, Lean
Animation	Blender, Unity

Interest

Synthetic Geometry	Study geometry without using analysis, including algebraic geometry, arithmetic geometry, algebraic analysis and more
Homotopy	The geometry up to "deformation", like A1 homotopy theory, homotopy type theory
Higher Structures	Higher means add homotopy to everything, like set, algebra, category, topos, TQFT, etc.